tec-speed 20.0 VTM1000i

UL Approval: E214381 Version:20/11/2024

PROCESS GUIDE

Precautions in Handling

Storage Condition & Shelf Life

		Laminate
Storage Condition	Temperature	Room
	Relative Humidity	/
Shelf Life		24 Months(airproof)

Designing and Inner layer Process

- Tooling & pinning holes should be drilled or routed. Punching is forbidden.
- Dimension stability is superior to Standard FR-4 materials.
- Please consult with your oxide vendor to make sure that our material is fully compatible with your oxide process. We recommend controlling the peel strength to over 2 lb./in.
- After oxide, it is preferred to press cores within 12 hours. Any cores staged more than 24 hours should be re-treated.
- Please confirm the grinding treatment process as the material was soft and brittle. Please don't use the grinding process for the inner and outer layer with large area or full no copper design.

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Availability-Laminate

Laminate	Core Thickness (inch)	Dk@10GHz	Df@10GHz
		IPC-TM650 2.5.5.5 Clamped Stripline	Cavity resonator
VTM1000i	0.015	9.8±0.245	0.0023
	0.020	9.8±0.245	0.0023
	0.025	9.8±0.245	0.0023
	0.030	9.8±0.245	0.0023
	0.050	9.8±0.245	0.0023
	0.060	9.8±0.245	0.0023
	0.075	9.8±0.245	0.0023
	0.100	9.8±0.245	0.0023
	0.125	9.8±0.245	0.0023
	0.150	9.8±0.245	0.0023

Remarks:

- Copper foil: Hoz~2oz, HTE, RTF and HVLP are all available based upon request.
- More types may be available upon request.
- Metal base is available upon request.

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Typical Drilling Parameters

- Drilling parameters should be adjusted depending on hole size, layer count, panel thickness, stack count and stack height etc.
- Please adjust drilling parameters after checking qualities of through holes.

Suggest Drilling parameter as blow:

Diameter (mm)	Spindle speed (KRPM)	Feed rate (IPM)	Retract (IPM)	Hit Counts
0.50	55	150	1000	500
0.60	46	130	1000	500
1.00	30	86	1000	500
2.53	15	45	1000	500

- Chip load: 0.0025”~0.005” / revolution.
- Surface speed: 300~400 SFM.
- Excessive wear of carbide drill bits. Diamond coated drill bits are recommended.
- LEAluminum sheet and white phenolic entry board are preferred, please consult your local Ventec representative for options.

Desmear Process

Panels should be baked for at least 1 hour at 120°C prior to plasma treatment. Plasma treated holes are more delicate than permanganate etched holes. Panels should not be exposed to any pressure wash or scrubbing process prior to metallization.

Plasma Desmear:
One time of FR-4 conditions is recommendable.

Permanganate horizontal Desmear:
The weight loss of VTM1000i laminate is more than that of our conventional FR-4 material. One time of FR-4 condition is recommendable.

Permanganate vertical Desmear:
Vertical desmear process is not recommendable.

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Routing

Consists of a hydrocarbon highly filled with ceramic filler. Due to the abrasive nature of the ceramic filler, the routing parameters and tool life should be adjusted accordingly.

Recommended Tools	Diamond-coated router bits
Lateral Chip Load	0.0010" -0.0015"
Surface Speed	200-400 SFM
Entry board	Phoenolic (0.010"-0.030")
Exit board	Phoenolic (0.100")

Packaging and baking recommendation

- It is recommended to bake the board before packaging at 125°C/4~8h to avoid moisture causing a decrease in heat resistance.
- If the PCBs needs to be stored for a long time before use, it is recommended to use aluminum foil vacuum packaging.
- If exceed 3 months after packaging, It is best to bake the PCBs at 125°C/4~6h before assembly before use.